

Sampling

Signals and Linear Systems

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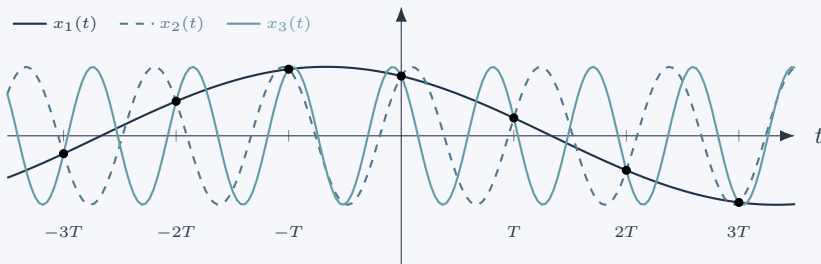
- **Introduction:** representing a continuous-time signal by its samples — and the ambiguity
- **Sampling as multiplication:** the time-domain row of the map
- **Two roads to $X_p(j\omega)$:** multiply-then-transform (a dead end) versus transform-then-convolve
- **The tools:** duality, multiplication $\leftrightarrow$ convolution, the spectrum of an impulse train
- **The key observation:** convolving with impulses copies the spectrum
- **The Sampling Theorem:** Nyquist rate and aliasing
- **Signal Reconstruction:** ideal low-pass filtering, zero-order hold, linear interpolation
- **Revisiting the DFT:** practical considerations and periodic signals

Introduction and Terminology

- In Lecture 1 we turned a continuous-time signal $x(t)$ into a sequence by reading it at uniform intervals, $x[n] = x(nT)$, and never looked back.
- **Question:** under what conditions is $x(t)$ uniquely determined by — and exactly recoverable from — its samples?
- Exact recovery is possible if the signal is **band-limited** (its Fourier transform is zero outside a finite band of frequencies) and the samples are taken sufficiently close together.
- This is the **Nyquist–Shannon Sampling Theorem**. Today we derive it, and the derivation is one picture.

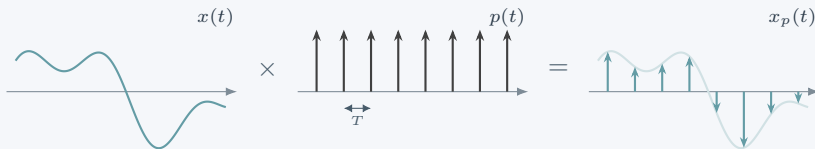
Ambiguity in Representation by Samples

- Generally, infinitely many continuous-time signals can yield the exactly same discrete sequence when sampled at interval T .



Sampling as Multiplication: The Time-Domain Row

- To analyze sampling mathematically, represent it as multiplying $x(t)$ by a periodic impulse train $p(t)$. (Lecture 1 wrote the period as T_s ; here it is simply T .)



Impulse-Train Sampling

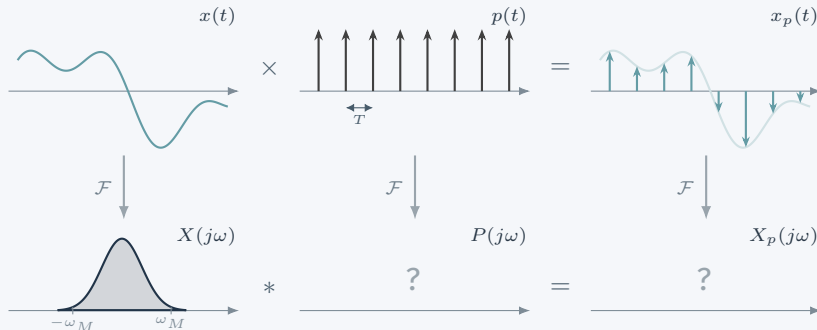
$$p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT), \quad x_p(t) = x(t)p(t) = \sum_{n=-\infty}^{+\infty} x(nT) \delta(t - nT)$$

- Each impulse picks up the value of $x(t)$ where it stands, so the weights $x(nT)$ are exactly the samples $x[n]$. So far, this is the picture from Lecture 1.

Two Roads to $X_p(j\omega)$

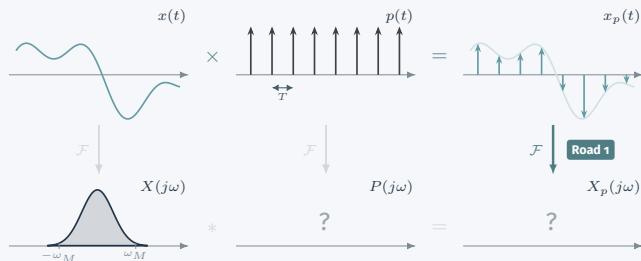
The Map: What We Know and What We Want

- Suppose we know $x(t)$ — and therefore its spectrum $X(j\omega)$. Sampling hands us $x_p(t)$. What is its spectrum $X_p(j\omega)$, and how is it related to $X(j\omega)$?



- Top row: known. Bottom row: two question marks. From the top-left corner to the bottom-right one there are two roads.

Road 1: Multiply First, Then Transform



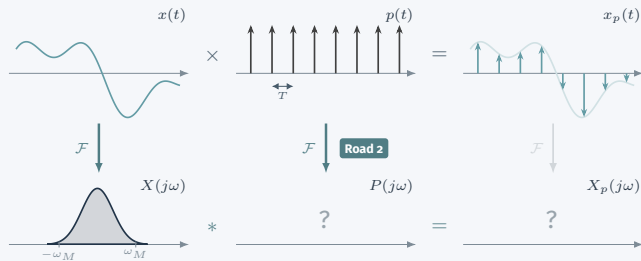
The product is already in hand, so just transform it; the impulses sift the integral:

$$X_p(j\omega) = \int_{-\infty}^{\infty} \sum_n x(nT) \delta(t - nT) e^{-j\omega t} dt = \sum_{n=-\infty}^{\infty} x(nT) e^{-j\omega nT}$$

A formula, not a picture

✓ Correct — Lecture 1 relabelled $\omega T \mapsto \omega$ and called it the DTFT. ✗ It says what $X_p(j\omega)$ is, not how it looks. ✗ $X(j\omega)$, the one thing we were given, appears nowhere. **It cannot be interpreted.** Park it.

Road 2: Transform First, Then Convolve



- Go down first: $X(j\omega)$ we have, $P(j\omega)$ we can look for. Then combine along the bottom row.
- A product in time should become a convolution in frequency, $X_p(j\omega) = \frac{1}{2\pi} X(j\omega) * P(j\omega)$. If so, $X_p(j\omega)$ is built out of $X(j\omega)$ – and then maybe, just maybe, we can read off how the two spectra are related.

Two tools to build first

- (1) Multiplication $\leftrightarrow$ convolution, which we get from the convolution property by **duality**.
- (2) The spectrum $P(j\omega)$ of the impulse train.

- We derive the **duality principle** from the inverse Fourier transform:

$$x(t) = \frac{1}{2\pi} \int X(j\omega) e^{j\omega t} d\omega \implies 2\pi x(-t) = \int X(j\omega) e^{-j\omega t} d\omega$$

- Swapping the names t and ω :

$$2\pi x(-\omega) = \int X(jt) e^{-j\omega t} dt = \mathcal{F}\{X(jt)\}$$

- Hence, the duality principle:

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega) \implies X(jt) \xleftrightarrow{\mathcal{F}} 2\pi x(-\omega)$$

- Every transform pair we know comes with a free twin: swap the roles of time and frequency, flip, and scale by 2π .

Tool 2: Multiplication Is Convolution in the Frequency Domain

- We know from the Convolution Property that,

$$x_1(t) * x_2(t) \xleftrightarrow{\mathcal{F}} X_1(j\omega)X_2(j\omega)$$

- By duality,

$$X_1(jt) * X_2(jt) \xleftrightarrow{\mathcal{F}} 2\pi x_1(-\omega)x_2(-\omega)$$

$$X_1(jt) \xleftrightarrow{\mathcal{F}} 2\pi x_1(-\omega)$$

$$X_2(jt) \xleftrightarrow{\mathcal{F}} 2\pi x_2(-\omega)$$

- Substituting $X_1(jt)$ with $u(t)$ and $X_2(jt)$ with $v(t)$:

$$u(t) \xleftrightarrow{\mathcal{F}} 2\pi x_1(-\omega) = U(j\omega)$$

$$v(t) \xleftrightarrow{\mathcal{F}} 2\pi x_2(-\omega) = V(j\omega)$$

- Substituting we get:

$$u(t)v(t) \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} U(j\omega) * V(j\omega)$$

- Hence the bottom row of the map really is a convolution — one unknown left, $P(j\omega)$:

$$X_p(j\omega) = \frac{1}{2\pi} [X(j\omega) * P(j\omega)]$$

Tool 3, First Attempt: $P(j\omega)$ Straight from the Definition

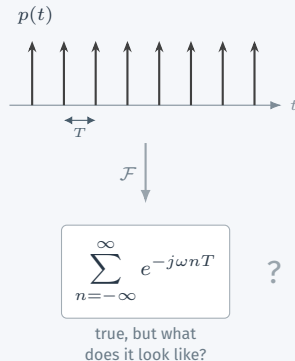
- Transform the train term by term; each impulse sifts:

$$P(j\omega) = \int_{-\infty}^{\infty} \sum_n \delta(t - nT) e^{-j\omega t} dt = \sum_{n=-\infty}^{\infty} e^{-j\omega nT}$$

- The same kind of answer as Road 1, with the same problem: an infinite sum of complex exponentials in ω . What does it look like? Worse, we would have to convolve $X(j\omega)$ with it. $\times$ Dead end #2.

Change of plan

$p(t)$ is **periodic** with period T , and periodic signals have a second representation: the **Fourier series**. Rewrite $p(t)$ as a Fourier series first, then transform. Be patient — in two slides it will be clear why this helps.



- $p(t)$ has period T , so with $\omega_s = 2\pi/T$:

$$p(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_s t}, \quad c_k = \frac{1}{T} \int_{-T/2}^{T/2} p(t) e^{-jk\omega_s t} dt$$

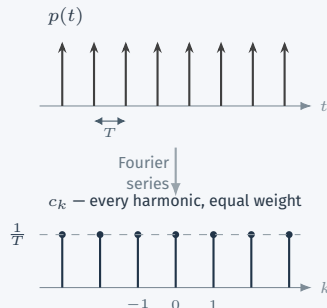
- Only one impulse, $\delta(t)$, lives inside one period. Sifting:

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_s t} dt = \frac{1}{T} e^0 = \frac{1}{T} \quad \text{for every } k$$

The impulse train as a Fourier series

$$p(t) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} e^{jk\omega_s t}$$

- A sum of exponentials again — but a very different one. Before: $e^{-j\omega nT}$, functions of ω we could not draw. Now: $e^{jk\omega_s t}$, **pure tones** in t at the frequencies $k\omega_s$. And the CTFT of a pure tone is the one spectrum everybody can draw.



The CTFT of a Pure Tone Is an Impulse

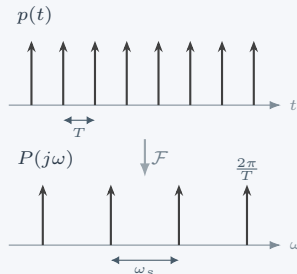
- Sifting gives $\delta(t) \leftrightarrow 1$. Duality with $x = \delta$, $X = 1$:

$$1 \xleftrightarrow{\mathcal{F}} 2\pi \delta(-\omega) = 2\pi \delta(\omega)$$

- Frequency shift, straight from the definition:

$$e^{j\omega_0 t} x(t) \xleftrightarrow{\mathcal{F}} \int x(t) e^{-j(\omega - \omega_0)t} dt = X(j(\omega - \omega_0))$$

- Hence a pure tone is a single impulse,
 $e^{jk\omega_s t} \leftrightarrow 2\pi \delta(\omega - k\omega_s)$, and the Fourier series transforms
term by term:



Spectrum of the impulse train

$$P(j\omega) = \frac{1}{T} \sum_k 2\pi \delta(\omega - k\omega_s) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s)$$

- Good news.** An impulse train in time is an impulse train in frequency, spaced $\omega_s = 2\pi/T$ apart (closer samples, sparser spectrum) – and impulses are the easiest thing in the world to convolve with.

Why Impulses Are Good News: What Convolution Does

- Recall where convolution came from: an LTI system with impulse response $h(t)$. Feed it $\delta(t)$, get $h(t)$. Feed it $\delta(t - t_0)$, get $h(t - t_0)$ — time invariance.

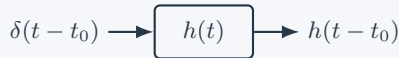
- Output = input * impulse response, so

$$\delta(t - t_0) * h(t) = h(t - t_0)$$

- Convolving with a shifted impulse does nothing but **copy** h to t_0 , scaled by the impulse's weight. By linearity, convolving with a train of impulses puts one copy at every impulse.

- The same in frequency, by sifting:

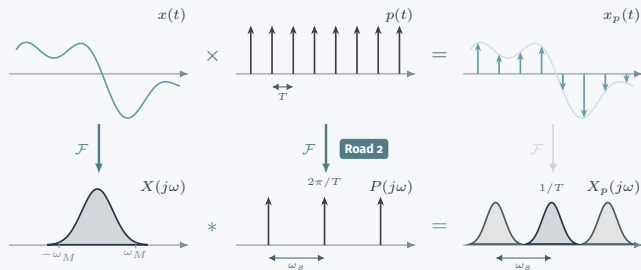
$$\begin{aligned} X(j\omega) * \delta(\omega - \omega_0) &= \int X(j\theta) \delta(\omega - \theta - \omega_0) d\theta \\ &= X(j(\omega - \omega_0)) \end{aligned}$$



Convolving with impulses just places copies



Back to the Map: Every Impulse Becomes a Copy of $X(j\omega)$



$$X_p(j\omega) = \frac{1}{2\pi} X(j\omega) * \frac{2\pi}{T} \sum_k \delta(\omega - k\omega_s) = \frac{1}{T} \sum_k X(j\omega) * \delta(\omega - k\omega_s) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X(j(\omega - k\omega_s))$$

Spectrum of the sampled signal

Copies of $X(j\omega)$, one at every impulse of $P(j\omega)$: spaced ω_s apart, each scaled by $1/T$. This is **the** observation behind the Nyquist–Shannon theorem. The copies sit ω_s apart and each is $2\omega_M$ wide — do they overlap?

The Sampling Theorem

- Suppose $x(t)$ is **band-limited**: $X(j\omega) = 0$ for $|\omega| > \omega_M$.
- On the map, $X_p(j\omega)$ is a row of copies of $X(j\omega)$ centred at $0, \pm\omega_s, \pm2\omega_s, \dots$; the copy at $k\omega_s$ occupies $[k\omega_s - \omega_M, k\omega_s + \omega_M]$.
- Neighbouring copies stay apart exactly when the right edge of one, ω_M , comes before the left edge of the next, $\omega_s - \omega_M$:

$$\omega_M < \omega_s - \omega_M \iff \omega_s > 2\omega_M$$

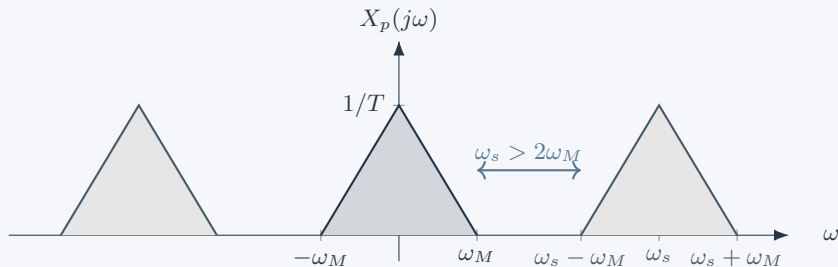
- If they stay apart, the copy at the origin is an untouched $X(j\omega)/T$, and a low-pass filter can cut it out: $x(t)$ is recoverable from its samples.
- If they overlap, the copies add where they meet, and no filter can un-mix them.

Nyquist-Shannon Sampling Theorem

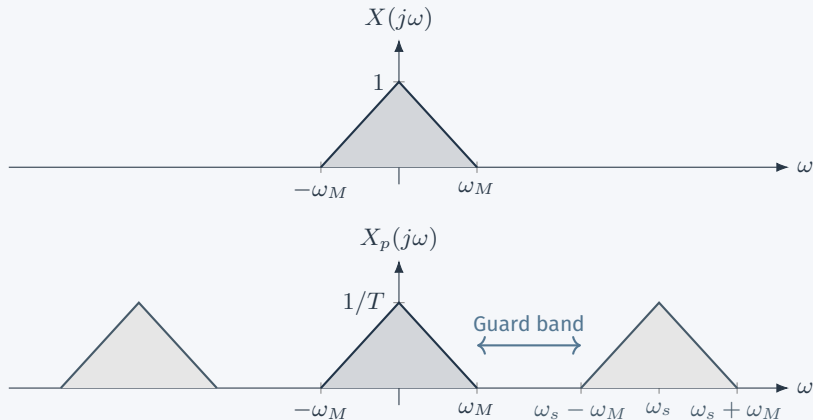
Let $x(t)$ be a band-limited signal with $X(j\omega) = 0$ for $|\omega| > \omega_M$. Then $x(t)$ is uniquely determined by its samples $x(nT)$, $n = 0, \pm 1, \pm 2, \dots$, if

$$\omega_s > 2\omega_M$$

where $\omega_s = 2\pi/T$. The minimum sampling frequency $2\omega_M$ is called the **Nyquist rate**.

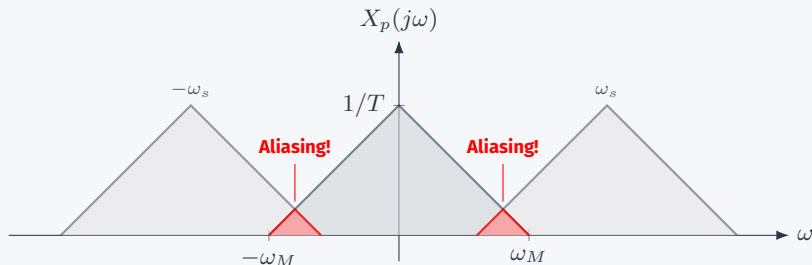


Visualizing the Spectrum: Oversampling



- When $\omega_s > 2\omega_M$, exact copies of the original spectrum are perfectly separated.
- The original signal can be recovered by isolating the baseband copy using a low-pass filter!

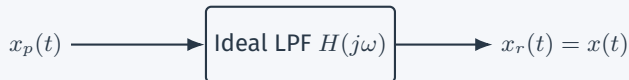
Visualizing the Spectrum: Undersampling (Aliasing)



- When $\omega_s < 2\omega_M$, the shifted copies overlap with the baseband spectrum.
- High frequencies fold over into the low-frequency band, distorting the signal. This phenomenon is called **aliasing**.
- The original signal $x(t)$ cannot be recovered by low-pass filtering.

Signal Reconstruction

- By the sampling theorem, if $\omega_s > 2\omega_M$, the **baseband spectrum** of $x_p(t)$ (the copy of the spectrum $X(j\omega)$ at the origin) is an exact replica of $X(j\omega)$ scaled by $1/T$.
- We can recover $x(t)$ by applying an **ideal low-pass filter** to $x_p(t)$.

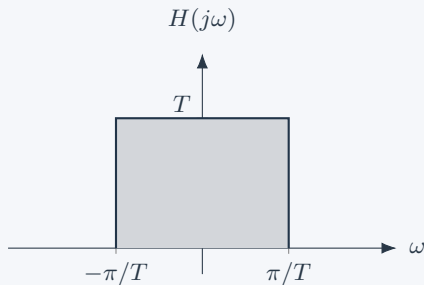


- The frequency response of the ideal LPF must be:

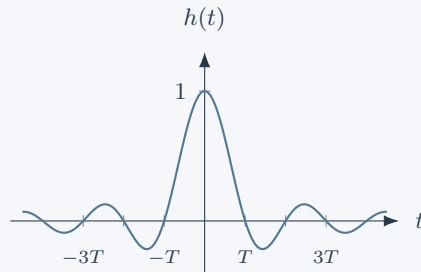
$$H(j\omega) = \begin{cases} T & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases} \quad \text{where} \quad \omega_M < \omega_c < \omega_s - \omega_M$$

- Commonly, we choose the cutoff frequency exactly in the middle: $\omega_c = \omega_s/2$, because it follows the assumption that $\omega_M < \omega_s/2$.

The Ideal Low-Pass Filter



Frequency Response $H(j\omega)$



Impulse Response $h(t)$

- The frequency response $H(j\omega)$ is a rectangular pulse with cutoff $\omega_c = \omega_s/2 = 2\pi f_s/2 = \pi/T$.
- The time-domain impulse response $h(t) = \text{sinc}(t/T)$ extends infinitely in both directions, making the ideal LPF **unrealizable**.

To find the time-domain impulse response $h(t)$ of an ideal low-pass filter with cutoff $\omega_c = \pi/T$ and constant amplitude T , we compute the Inverse Fourier Transform:

$$\begin{aligned} h(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H(j\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\pi/T}^{\pi/T} T e^{j\omega t} d\omega \\ &= \frac{T}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-\pi/T}^{\pi/T} = \frac{T}{\pi t} \left(\frac{e^{j(\pi/T)t} - e^{-j(\pi/T)t}}{2j} \right) \\ &= \frac{T}{\pi t} \sin\left(\frac{\pi t}{T}\right) = \text{sinc}\left(\frac{t}{T}\right) \quad \left[\text{Recall } \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x} \right] \end{aligned}$$

Result: The time-domain filter corresponding to a perfect frequency cutoff is exactly the infinitely wide sinc function!

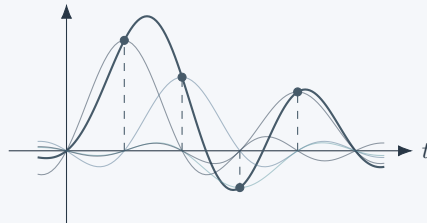
- The impulse response $h(t)$ of the ideal LPF (cutoff $\omega_c = \pi/T$) is:

$$h(t) = \text{sinc}\left(\frac{t}{T}\right) = \frac{\sin(\pi t/T)}{\pi t/T}$$

- The reconstructed signal $x_r(t) = x_p(t) * h(t)$ becomes:

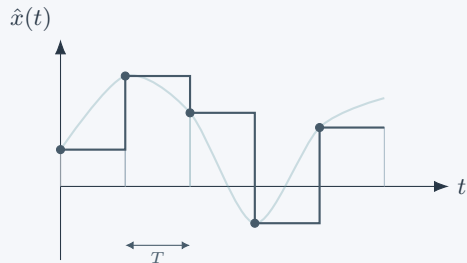
$$\begin{aligned} x_r(t) &= \left(\sum_{n=-\infty}^{+\infty} x(nT)\delta(t - nT) \right) * h(t) \\ &= \sum_{n=-\infty}^{+\infty} x(nT)\text{sinc}\left(\frac{t - nT}{T}\right) \end{aligned}$$

- Superposition of shifted sinc functions.
- However**, $h(t)$ is non-causal and decays very slowly. It is **unrealizable** in practice. We need approximations.



The simplest practical reconstruction method is the **Zero-Order Hold (ZOH)**.

- **Concept:** Simply hold the sample value constant until the next sample arrives.
- This forms a “staircase” approximation $x_0(t)$ of the continuous signal.
- Widely used in Digital-to-Analog Converters (DACs).



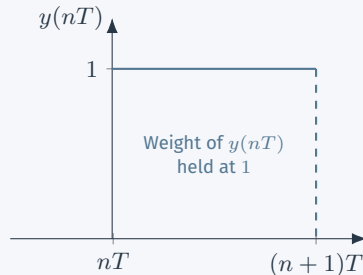
- We can view this mathematically as filtering the impulse-sampled signal $x_p(t)$ with a rectangular pulse $h_0(t)$.

Why is $h_0(t)$ a Rectangle?

- In a reconstructed output signal, an original sample $y(nT)$ contributes strictly via a shifted, scaled version of the base filter: $y(nT) \cdot h(t - nT)$.
- The shape of $h(t)$ dictates the **weighting** of the sample $y(nT)$ over continuous time.

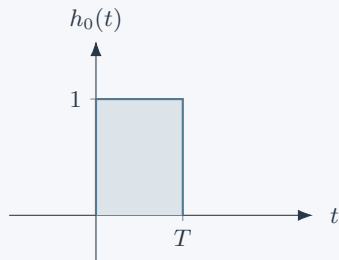
For Zero-Order Hold:

- We desire the sample $y(nT)$ to simply hold its full weight of 1 constant precisely until the next sample at $(n + 1)T$ arrives.
- Thus, the isolated weight of $y(nT)$ must jump to 1 at $t = nT$ and abruptly drop back to 0 at $t = (n + 1)T$.
- Therefore, the underlying impulse response $h_0(t)$ **must** inherently be a rectangular block formally spanning from 0 to T !

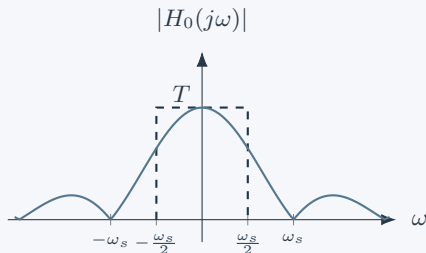


The ZOH impulse response $h_0(t)$ is simply a rectangular pulse equal to 1 over $[0, T]$. We construct $H_0(j\omega)$ strictly via continuous Fourier integration:

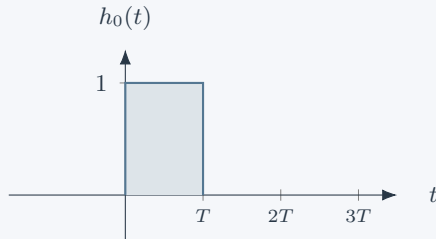
$$\begin{aligned} H_0(j\omega) &= \int_{-\infty}^{\infty} h_0(t) e^{-j\omega t} dt \\ &= \int_0^T (1) e^{-j\omega t} dt = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^T \\ &= \frac{1 - e^{-j\omega T}}{j\omega} = \frac{e^{-j\omega T/2} (e^{j\omega T/2} - e^{-j\omega T/2})}{j\omega} \\ &= e^{-j\omega T/2} \left[\frac{2}{\omega} \sin \left(\frac{\omega T}{2} \right) \right] \\ &= T e^{-j\omega T/2} \operatorname{sinc} \left(\frac{\omega T}{2\pi} \right) \end{aligned}$$



Frequency Domain view of ZOH



Frequency Response $|H_0(j\omega)|$



Impulse Response $h_0(t)$

- $h_0(t)$ is a rectangle of width T . Its frequency response includes a sinc function:

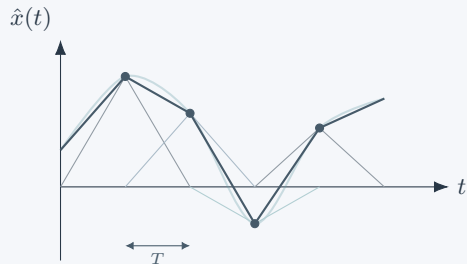
$$H_0(j\omega) = T e^{-j\omega T/2} \text{sinc}\left(\frac{\omega T}{2\pi}\right).$$

We notice that,

- In the passband $|\omega| \leq \omega_s/2$, the gain is not constant, attenuating high frequencies slightly.
- Outside the passband, the gain is non-zero, allowing high-frequency aliasing copies to leak through.

Linear Interpolation (First-Order Hold)

- **Linear Interpolation** connects adjacent samples with straight lines.
- This produces a continuous, much smoother approximation $\hat{x}_1(t)$ than the ZOH staircase.
- Achieved by filtering the impulse-sampled signal $x_p(t)$ with a triangular pulse $h_1(t)$.



- The impulse response $h_1(t)$ is a triangle spanning $[-T, T]$ with peak height 1 at $t = 0$.

Why is $h_1(t)$ a Triangle?

- In a reconstructed signal, a given sample $y(nT)$ contributes via $y(nT) \cdot h_1(t - nT)$.
- The shape of $h_1(t)$ thus dictates the **weighting** of $y(nT)$ over time.

For Linear Interpolation:

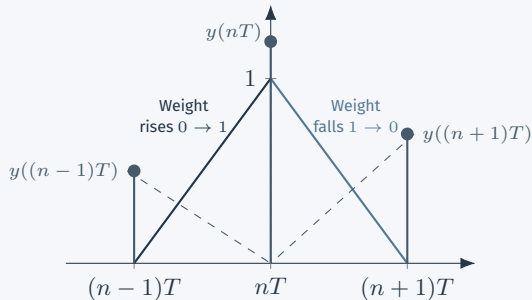
Given a weight $\alpha \in [0, 1]$, we should have,

$$y(t) = (1 - \alpha)y((n - 1)T) + \alpha y(nT) \quad \text{for } t \in [(n - 1)T, nT]$$

$$y(t) = (1 - \alpha)y(nT) + \alpha y((n + 1)T) \quad \text{for } t \in [nT, (n + 1)T]$$

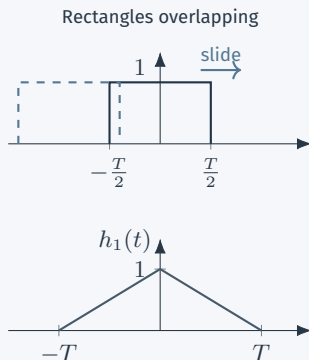
Therefore,

- The weight of $y(nT)$ should linearly rise strictly from 0 to 1 between $(n - 1)T$ and nT .
- It should linearly fall from 1 back to 0 between nT and $(n + 1)T$.
- At all other times, its weight must precisely be 0 to isolate its exact local influence.
- Therefore, $h_1(t)$ **must** inherently be a symmetric triangle!



A linear interpolation triangle filter $h_1(t)$ can be formed by convolving two shifted zero-order hold rectangles!

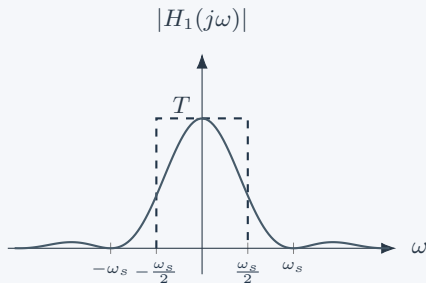
$$h_1(t) = \frac{1}{T} \left(h_0(t + T/2) * h_0(t + T/2) \right)$$



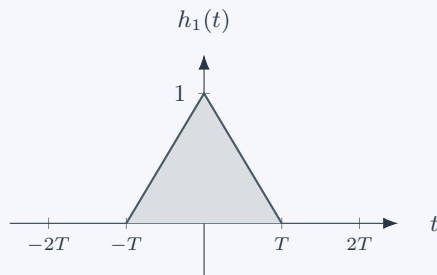
- $h_0(t + T/2)$ is just the standard ZOH block, centered exactly on time $t = 0$.
- Area of overlap grows linearly to a peak, and drops cleanly linearly!
- The peak area is $T \times 1 = T$, hence the factor $\frac{1}{T}$, as the triangle peak should be 1.

Frequency Derivation: Use $\mathcal{F}\{x * x\} = \mathcal{F}\{x\}^2$.

$$\begin{aligned} H_1(j\omega) &= \frac{1}{T} \left[\mathcal{F}\{h_0(t + T/2)\} \right]^2 \\ &= \frac{1}{T} \left[T \operatorname{sinc}\left(\frac{\omega T}{2\pi}\right) \right]^2 \\ &= T \operatorname{sinc}^2\left(\frac{\omega T}{2\pi}\right) \end{aligned}$$



Frequency Response $|H_1(j\omega)|$

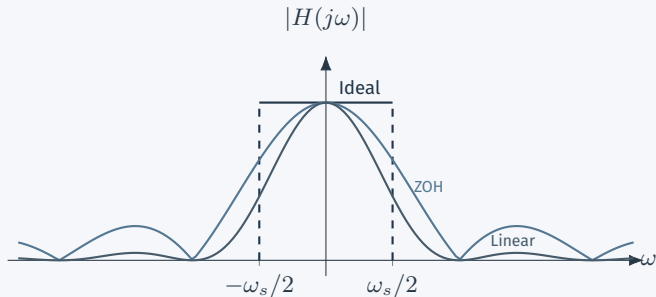


Impulse Response $h_1(t)$

- $h_1(t)$ is a triangular pulse spanning $[-T, T]$. Its frequency response is $H_1(j\omega) = T \text{sinc}^2\left(\frac{\omega T}{2\pi}\right)$.
- Because of the convolution property, $H_1(j\omega)$ is proportional to the square of the ZOH response, without the phase delay.

Comparing Reconstruction Filters

- Let's compare the frequency domain magnitudes against the ideal filter (assuming $T = 1$):



- Linear interpolation has much faster high-frequency rolloff (less leakage) than ZOH, making it a "better" low-pass filter approximation.

Connection to the DFT

The Uncertainty Principle

Signals **cannot** be both time-limited and strictly band-limited!

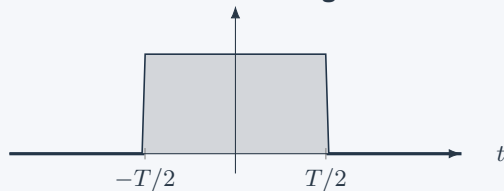
- Finite-duration captures mean $X(j\omega)$ inherently extends to infinite frequencies.^a

^aInteresting watch: 3Blue1Brown - Uncertainty Principle

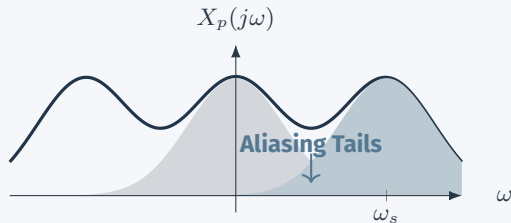
Unavoidable Aliasing

Sampling any real-world signal causes some aliasing! The infinite spectral tails will always overlap and leak into our baseband.

Time-Limited Signal



Infinite Frequency Spectrum & Aliasing



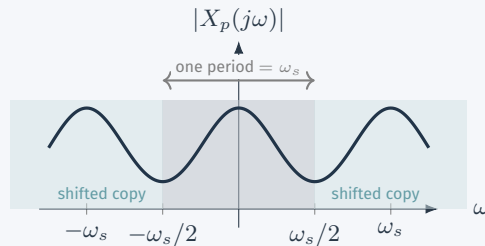
Road 1 gave us the CTFT of the sampled impulse train directly:

$$X_p(j\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega nT}$$

$$\begin{aligned} X_p(j(\omega + \omega_s)) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j(\omega + \omega_s)nT} \\ &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega nT} e^{-j2\pi n} \\ &= X_p(j\omega) \end{aligned}$$

- Road 2 says this same function is the train of copies $\frac{1}{T} \sum_k X(j(\omega - k\omega_s))$.
- It repeats every $\omega_s = \frac{2\pi}{T}$.
- Under $\omega T \mapsto \omega$, the same expression becomes the DTFT:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}.$$

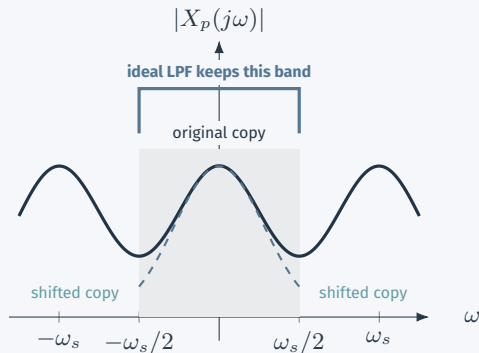


Baseband Copy, Aliasing, and the Ideal LPF

$$X_p(j\omega) = \frac{1}{T} \sum_{r=-\infty}^{\infty} X(j(\omega - r\omega_s))$$

- The central band $[-\omega_s/2, \omega_s/2]$ contains the original copy centered at 0.
- Every other interval of width ω_s is a shifted copy of that same spectrum.
- For finite-duration captures the tails are infinite, so some aliasing leakage remains, but the baseband picture is still the same one from the Nyquist proof.
- The ideal reconstruction filter keeps exactly that baseband copy:

$$H(j\omega) = \begin{cases} T, & |\omega| < \omega_s/2 \\ 0, & \text{otherwise} \end{cases}$$



- Ideal reconstruction uses one ω_s -wide period of the sampled spectrum:

$$x(t) = \frac{1}{2\pi} \int_{-\omega_s/2}^{\omega_s/2} X_p(j\omega) e^{j\omega t} T d\omega$$

- After the same mapping $\omega T \mapsto \omega$ used to derive the DTFT,

$$\begin{aligned} X_p(j\omega) &\longrightarrow X(e^{j\omega}) \\ T d\omega &\longrightarrow d\omega \end{aligned}$$

- Also,

$$[-\omega_s/2, \omega_s/2] \xrightarrow{\omega T \mapsto \omega} [-\pi, \pi]$$

Applying the mapping at the sample locations,

$$\begin{aligned} x[n] = x(nT) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\frac{\omega}{T} T n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \end{aligned}$$

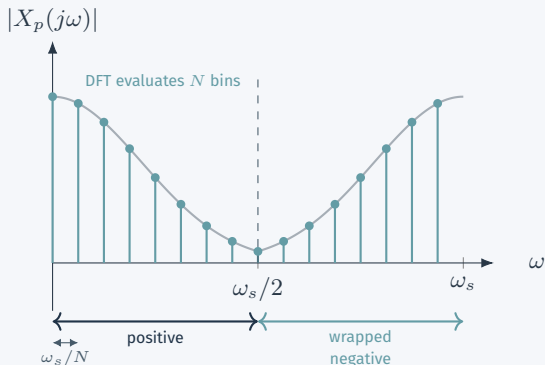
Inverse DTFT

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

As the area is the same regardless of which period we choose.

$$X[k] = X_p\left(j\frac{k\omega_s}{N}\right) = X(e^{j\omega})\Big|_{\omega=\frac{2\pi}{N}k}$$

- Undoing the relabeling, a normalized DTFT point at ω sits at the physical frequency $\omega/T = \omega\omega_s/(2\pi)$.
- So an N -point DFT takes N uniformly spaced samples across one physical period of width ω_s .
- For even N , bins $k = 0, \dots, N/2$ run from dc to the highest positive frequency.
- Bins $k = N/2 + 1, \dots, N - 1$ then continue as wrapped negative frequencies: large negative first, then back toward dc.



Periodic Signals Give Harmonic Line Spectra

- If the underlying signal has period T_0 , then $\omega_0 = 2\pi/T_0$ and

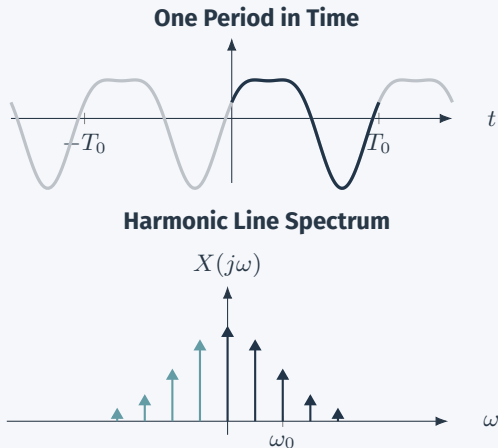
$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}.$$

- Its CTFT is therefore a line spectrum:

$$X(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0).$$

- If we sample exactly N times in one period, then

$$T_0 = NT, \quad \omega_s = \frac{2\pi}{T} = N\omega_0.$$

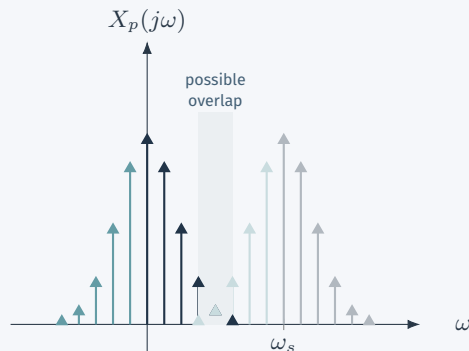


Sampling Copies the Periodic Line Spectrum

From the Nyquist proof, the sampled spectrum is a sum of shifted copies of the original spectrum:

$$\begin{aligned}X_p(j\omega) &= \frac{1}{T} \sum_{r=-\infty}^{\infty} X(j(\omega - r\omega_s)) \\&= \frac{1}{T} \sum_{r=-\infty}^{\infty} 2\pi \sum_{k=-\infty}^{\infty} a_k \delta((\omega - r\omega_s) - k\omega_0) \\&= \frac{2\pi}{T} \sum_{r=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0 - r\omega_s)\end{aligned}$$

- Each harmonic line is copied to every passband.
- Depending on the bandwidth, the copied line spectra may overlap and alias into each other.



Nyquist Condition for the Periodic Case

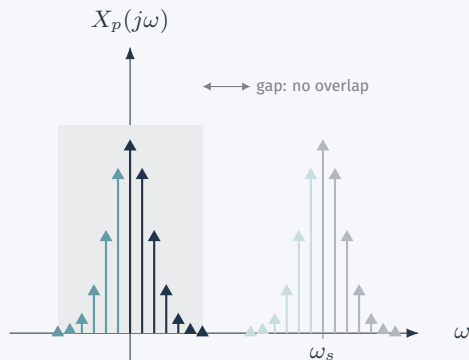
If the original periodic signal is bandlimited to the Nyquist band,

$$|k\omega_0| < \omega_s/2,$$

then $a_k = 0$ whenever $|k| \geq N/2$. The copied line spectrum becomes

$$X_p(j\omega) = \frac{2\pi}{T} \sum_{r=-\infty}^{\infty} \sum_{|k| < N/2} a_k \delta(\omega - k\omega_0 - r\omega_s).$$

- Now one passband contains one clean, non-overlapping copy of the original harmonic line spectrum.
- This is the periodic-signal version of the Nyquist condition.



The Same Spectrum from the Sampled Sequence

From Lecture 1 (DTFT and DFT), the sampled periodic signal itself has the Fourier series

$$x_p(t) = \sum_{\ell=-\infty}^{\infty} c_{\ell} e^{j\ell\omega_0 t}, \quad c_{\ell} = \frac{1}{T_0} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}\ell n} = \frac{1}{T_0} X[\ell].$$

Since $c_{\ell+N} = c_{\ell}$, choose any N consecutive indices $\mathcal{K}$ and write $\ell = k + rN$:

$$\begin{aligned} X_p(j\omega) &= 2\pi \sum_{\ell=-\infty}^{\infty} c_{\ell} \delta(\omega - \ell\omega_0) \\ &= 2\pi \sum_{r=-\infty}^{\infty} \sum_{k \in \mathcal{K}} c_{k+rN} \delta(\omega - (k+rN)\omega_0) \\ &= 2\pi \sum_{r=-\infty}^{\infty} \sum_{k \in \mathcal{K}} c_k \delta(\omega - k\omega_0 - r\omega_s). \end{aligned}$$

This equation matches exactly with the previous one. Therefore, the coefficients must be related.

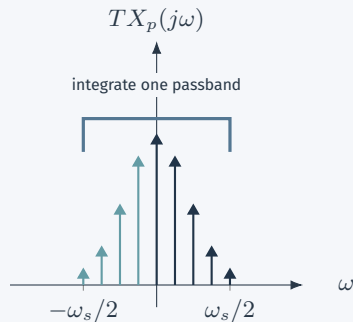


Coefficient Match

$$c_k = \frac{a_k}{T}, \quad X[k] = T_0 c_k = \frac{T_0}{T} a_k = N a_k$$

Hence, if the periodic signal is bandlimited to the Nyquist band, the DFT bins are scaled versions of the original Fourier series coefficients.

Reconstruction Gives $x(t)$ and the IDFT



Inside one passband only the non-overlapping harmonics remain:

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\omega_s/2}^{\omega_s/2} TX_p(j\omega) e^{j\omega t} d\omega \\ &= T \sum_{|k| < N/2} c_k e^{jk\omega_0 t} = \sum_{|k| < N/2} a_k e^{jk\omega_0 t}. \end{aligned}$$

So the LPF recovers $x(t)$ exactly when Nyquist holds. With $t = nT$, $T_0 = NT$:

$$\begin{aligned} x[n] = x(nT) &= T \sum_{|k| < N/2} c_k e^{jk\omega_0 nT} \\ &= \frac{1}{N} \sum_{|k| < N/2} X[k] e^{j\frac{2\pi}{N} kn}. \end{aligned}$$

Periodicity of $X[k]$ then shifts the index set to $k = 0, \dots, N-1$:

Inverse DFT

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N} kn}$$